

Deep Learning for High-Fidelity Document Processing

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Abstract: In this paper, we propose a novel neural architecture for document conversion. We demonstrate state-of-the-art accuracy on multi-column layouts, mathematical equations, and complex tables.

I. INTRODUCTION

Document conversion from fixed-layout PDF to flowing formats has long been a challenging problem in computer science. Scientific papers contain complex structural elements including formulas.

II. ALGORITHM & METHOD

Algorithm 1: Layout-Aware Reflow
Input: Bounding boxes B , Text spans S
Output: Document hierarchy H
1: `def process_columns(spans):`
2: `columns = cluster_x(spans)`
3: `return sort_reading_order(columns)`

$$E = \sum_{i=1}^n x_i^2 + \int_0^{\infty} f(t) dt$$

Model Name	BLEU Score	Layout Accuracy	Time (s)
Baseline Converter	72.4	81.2%	1.84
OmniPDF (Ours)	98.7	99.5%	0.42

Table 1: Benchmark evaluation on scientific paper dataset.

REFERENCES

- [1] J. Doe, 'Neural Document Parsing', IEEE Trans. PAMI, 2024.
[2] A. Smith, 'Layout Preservation in Word Documents', ACM SIGGRAPH, 2025.